

Multiple Choice

1. **Answer:** A

Options: A) Sensory memories; B) Personal concerns; C) Marital strain; D) Job satisfaction

Note: Sensory memories are brief and automatic recordings of sensory information that do not significantly decline with age, thus remaining relatively stable over the adult years.

2. **Answer:** C

Options: A) on the posterior wall of the oropharynx.; B) under the mucosa of the ventral surface of the tongue.; C) between the palatoglossal and palatopharyngeal folds.; D) at all three sites.

Note: Tonsillar tissue is located in the oropharynx, specifically between the palatoglossal and palatopharyngeal folds, where it forms the palatine tonsils, which are key components of the immune system in the throat.

3. **Answer:** A

Options: A) Intron; B) 3' Poly A tail; C) Ribosome binding site; D) 5' cap

Note: Introns are non-coding sequences within the initial RNA transcript that are spliced out during RNA processing before the mature mRNA is translated into protein.

4. **Answer:** C

Options: A) arousal; B) erection; C) inhibition; D) orgasm

Note: The correct answer is inhibition because the consumption of alcohol, even in small amounts like one beer, can lower inhibitions by affecting the central nervous system and impairing judgment and self-control.

5. **Answer:** B

Options: A) Active observer; B) Participant observer; C) Indiscreet observer; D) Meddling observer

Note: The term "participant observer" refers to a research method in which the researcher actively engages in the environment or community being studied, allowing for a deeper understanding of the social dynamics and behaviors from an insider's perspective.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: An egg can only remain viable for fertilization for about 12 to 24 hours after ovulation, making the statement false.

7. **Answer:** False

Options: A) True; B) False

Note: The statement is false because male hormones, primarily testosterone, are predominantly produced by the testes, not the prostate gland.

8. **Answer:** False

Options: A) True; B) False

Note: A dental panoramic radiograph provides a comprehensive view of the entire oral and maxillofacial region, including the hard palate, zygomatic arch, mandibular canal, and maxillary sinus, rather than excluding any of these structures.

9. **Answer:** False

Options: A) True; B) False

Note: The statement is false because individuals with an external locus of control tend to attribute their successes to external factors, such as luck or the actions of others, rather than to their own skills and efforts.

10. **Answer:** True

Options: A) True; B) False

Note: The statement is true because rigorous screening processes and testing protocols for blood donations in the United States have significantly reduced the risk of HIV transmission through blood transfusions to approximately 1 in 2 million units.

Long Form Response

11. **Answer:** Marfan syndrome is primarily caused by mutations in the FBN 1 gene, which encodes the protein fibrillin-1. Fibrillin-1 is a crucial component of the extracellular matrix, playing a significant role in the formation of microfibrils that provide structural support in connective tissues. Mutations in this gene lead to a deficiency or dysfunction of fibrillin-1, resulting in various phenotypic manifestations such as tall stature, long limbs, and cardiovascular issues, particularly aortic dilation and dissection. The implications of these mutations on health are profound, as affected individuals face an increased risk of life-threatening complications, necessitating regular medical monitoring and potential interventions. Thus, the genetic basis of Marfan syndrome underscores the critical role of fibrillin in maintaining connective tissue integrity and overall health.

Note: correct because it accurately identifies the FBN 1 gene's mutations as the genetic cause of Marfan syndrome, explains the role of fibrillin-1 in connective tissue structure, and describes the resulting phenotypic characteristics and health complications associated with the disorder.

12. **Answer:** Chromosome instability syndromes are characterized by an increased frequency of chromosomal abnormalities, such as structural defects and numerical variations, due to defects in DNA repair mechanisms. These syndromes often lead to a variety of health complications, including increased cancer risk and developmental issues. In contrast, Klinefelter syndrome, which results from the presence of an extra

X chromosome in males (47,XXY), does not exhibit the same instability. Instead, Klinefelter syndrome is primarily a chromosomal aneuploidy rather than a syndrome of instability, as it is characterized by specific phenotypic traits such as infertility, gynecomastia, and learning difficulties without the widespread chromosomal aberrations that typify instability syndromes. Therefore, Klinefelter syndrome does not fit the classification of chromosome instability syndromes due to its distinct genetic basis and clinical presentation.

Note: correct because chromosome instability syndromes involve defects in DNA repair leading to frequent chromosomal abnormalities, while Klinefelter syndrome results from a stable chromosomal anomaly (an extra X chromosome) without the associated instability or increased cancer risk typical of instability syndromes.

End of Answer Key